Week 6 Lecture

Topic: Understanding Graphs and Short Talks

Part 1: Oral expression – Describing simple graphs and charts

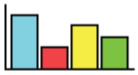
Graphs and Diagrams

A Types of diagrams

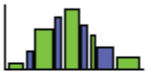
Diagrams are visual ways of **presenting data** concisely. They are often also called **figures**. In an academic article they are usually **labelled** Fig. (Figure) 1, Fig. 2, etc.



A **pie chart** is a circle divided into **segments** from the middle (like slices of a cake) to show how the total is divided up. A **key** or **legend** shows what each segment represents.



A **bar chart** is a diagram in which different amounts are represented by thin vertical or horizontal bars which have the same width but **vary** in height or length.



A **histogram** is a kind of bar chart but the bar width also varies to indicate different values.

Number	Amount
1	10
2	5
3	20

A **table** is a grid with **columns** and **rows** of numbers.



A **cross-section** is something, or a model of something, cut across the middle so that you can see the inside. A cross-section of the earth's crust, for example, shows the different **layers** that make it up. A **label** gives the name of each part of the cross-section. Cross-section can also be used to mean a small group that is representative of all the different types within the total group (e.g. *the survey looked at a cross-section of society*).

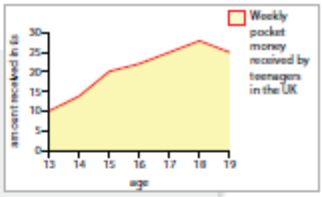
A **flowchart** is a diagram which indicates the **stages** of a process.



Common Mistake

Don't use the definite article (*the*) when referring to a specific diagram. *See Table 4 below.*
(NOT ~~See the table 4~~ …)

B A graph



The **graph presents** data relating to teenagers and pocket money. A **random sample** of 1,000 teenagers were surveyed and the average pocket money received at each age has been plotted on the graph. The **x axis** or **horizontal axis indicates** age and the **y axis** or **vertical axis shows** the amount of money received per week. The **graph shows** that 15-year-olds receive twice as much pocket money as 13-year-olds. **From the** graph **we can see** that the amount received **reaches a peak** at the age of 18 and then starts to decline. This **decline** can perhaps **be explained by the fact that** many teenagers start earning and stop receiving pocket money at the age of 18.

Graphs are drawn by **plotting** points on them and then drawing a line to join **adjacent** points. If there are two separate lines on a graph, the lines can **cross** or **intersect** at various points. Lines that **run parallel** to one another never intersect.

Graphs show how numbers **increase** or **decrease**. Numbers can also be said to **rise** or **grow** and **fall**, **drop** or **decline**. Other verbs used about growth include **double**[1], **soar**[2], **multiply**[3], **appreciate**[4] and **exceed**[5] [another number].

[1] increase to twice the number or amount; opposite = **halve** [2] increase very quickly and by a large amount; opposite = **plummet** [3] increase to a very large number [4] increase in value; opposite = **depreciate** [5] increase to greater than a particular number or amount; opposite = **fall below**

Language help

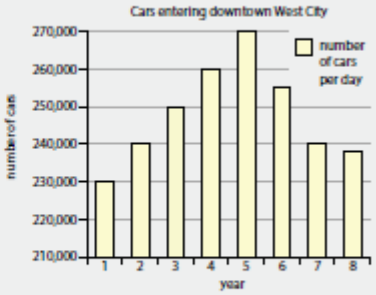
The verbs *increase* and *decrease* are followed by *by* (e.g. *The population of the city has increased by 10%.*). The nouns *increase, rise, growth, fall, drop* and *decline, decrease*, are followed by *in* (to explain what is rising) or *of* (to explain the size of the change), e.g. *a rise / an increase **of** 15% **in** the number of cars.*

Practice Activities:

Activity 1

Look at the chart and complete the text with the missing words.

Cars entering downtown West City



The chart [1]..... the number of cars entering the downtown area of West City each day over an eight-year period (years 1-8). The totals are listed on the [2]..... axis (give two answers), while the years are listed on the [3]..... axis (give two answers). To the top right of the graph we see the [4]..... .The number of cars [5]v..... over the period. The total rose in the first few years and [6]r..... a [7]p..... in year 5, after which the numbers started to [8]..... . This decline can be [9]..... by the [10]..... that a new mass transit railway was opened in year 6, which is a clear illustration of how good public transport can dramatically affect car use.

Activity 2

Answer the questions.

1 Draw examples of a pie chart and a bar chart.
2 What is the best type of diagram to present the different layers of rock in the Grand Canyon?
3 In a table, what is the difference between columns and rows?
4 What would be the best type of diagram to present the different stages in a research project you did?
5 How many segments are there in the pie chart in A opposite?
6 If you look at two adjacent columns in a table, are they next to each other or separated?
7 What is another name for a legend in a diagram?
8 What type of data collection are you doing if you survey the first 50 people you come across?
9 What do two lines on a graph do if (a) they intersect and (b) they run parallel to each other?
10 Choose the correct sentence: (a) There was an increase in 12% of smart phone sales last year.
(b) There was an increase of 12% in smart phone sales last year.

Activity 3

Replace the underlined words with more precise, academic words.

1 The different bits of the pie chart show the numbers of people in each age group.
2 She kept a record by marking the midday temperature on a graph for a month.
3 People's salaries usually reach their highest point when they are in their late 40s.
4 This flowchart shows the different bits of our project over the next five years.
5 The two lines on the graph cross each other at point A.
6 Draw a line connecting the points that are next to each other.
7 The government's popularity in the opinion polls is beginning to go down.
8 If you look along the third line of the table you can see the figures for the 1950s.

Activity 4

Rewrite the underlined words and phrases using words from B opposite. There is also a deliberate mistake in one of the sentences. Can you find it and correct it?

1 Populations of some bird species in South Asia have crashed by 97% in recent years. The number of cases of death by poisoning has increased sharply.
2 In 2007 the child mortality rate fell to lower than 60 deaths per 1,000.
3 The average family car in the UK goes down in value by 20% per year. This means its value has fallen by more than half after just three years.
4 A typical piece of land on the edge of the city will go up in value by 15% per year, and house prices have gone up rapidly by a large amount in the last six months.
5 Business courses have increased greatly in number while science programmes have gone down.
6 The temperature rose higher than 45°C in some parts of the country. See the figure 3.
7 Between 1983 and 2006, the number of this species of eagle went up from 22 pairs to 58. Other bird populations have gone up to twice the number in the same period.
8 The numbers of old soldiers attending regimental reunions are becoming smaller each year.

How to Describe Graphs and Charts in English

Line charts and bar charts have a vertical axis and a horizontal axis. Each axis is labeled to indicate what type of information it contains. Typical information included on vertical and horizontal axis include:

- age - how old
- weight - how heavy
- height - how tall
- date - which day, month, year, etc.
- time - how much time is required
- length - how long
- width - how wide
- degrees - how hot or cold
- percentage - a portion of 100%
- number - number
- duration - the length of time required

There are a number of specific words and phrases used to describe and discuss graphs and charts. This vocabulary is especially important when presenting to groups of people. Much of the language of graphs and charts relates to movement. In other words, the language of graphs and charts often speaks of small or large movement or differences between various data points. Refer to this language of graphs and charts to help improve your ability to speak about graphs and charts.

The following list the verb and noun used to speak about positive and negative movements, as well as predictions. Example sentences are found after each section.

Positive

- to climb - a climb
- to ascend - an ascent
- to rise - a rise
- to improve - an improvement
- to recover - a recover
- to increase - an increase
- *Sales have climbed over the past two quarters.*
- *We've experienced a rise in consumer demand.*
- *Consumer confidence recovered in the second quarter.*
- *There has been an increase of 23% since June.*
- *Have you seen any improvement in customer satisfaction?*

Negative

- to fall - a fall
- to decline - a decline
- to plunge - a plunge
- to decrease - a decrease
- to worsen - a slip
- to deteriorate - a dip

- *Research and development spending has fallen by 30% since January.*
- *Unfortunately, we've seen a decline over the past three months.*
- *As you can see, sales have plunged in the northwest region.*
- *Government spending has decreased by 10% over the past two years.*
- *There's been a slip in profits this past quarter.*
- *Comedy book sales have deteriorated for three quarters.*

Predicting Future Movement

- to project - a projection
- to forecast - a forecast
- to predict - a prediction
- *We project improved sales in the coming months.*
- *As you can see from the chart, we forecast increased research and development spending next year.*
- *We predict improving sales through June.*

This list provides adjectives and adverbs used to describe how quickly, slowly, extremely, etc. something moves. Each adjective/adverb pair includes a definition and an example sentence.

- slight - slightly = insignificant
- *There's been a slight decline in sales.*
- *Sales have declined slightly over the past two months.*
- sharp - sharply = quick, large movement
- *Investment rose sharply during the first quarter.*
- *We made a sharp increase in investment.*

- abrupt - abruptly = sudden change
- *Sales dropped abruptly in March.*
- *There was an abrupt drop in sales in March.*
- rapid - rapidly = quick, very fast
- *We expanded rapidly throughout Canada.*
- *The company made a rapid expansion throughout Canada.*
- sudden - suddenly = without warning
- *Unfortunately, consumer interest suddenly decreased.*
- *There was a sudden decrease in consumer interest in January.*
- dramatic - dramatically = extreme, very big
- *We've dramatically improved customer satisfaction over the past six months.*
- *As you can see from the chart, the dramatic growth has come after we invested in a new product line.*
- calm - calmly = evenly, without much change
- *The markets have reacted calmly to recent developments.*
- *As you can see from the graph, consumers have been calm over the past few months.*
- flat = without change
- *Profit has been flat over the past two years.*
- steady - steadily = no change
- *There has been a steady improvement over the past three months.*
- *Sales have improved steadily since March.*

Practice Activity:

Student A and student B – go to https://www.statista.com/, choose a graph, and describe it using the vocabulary from this section.

TED Talk: How to Spot a Misleading Graph

Listening Comprehension: Listening for gist

1. What is the main topic of the video?
   a) How to make graphs using computer software
   b) How graphs can mislead people, and how to spot it
   c) Why politicians and advertisers should not use numbers
   d) The history of statistics in advertising
2. What does the speaker say about the way people usually think of graphs?
   a) They are too complicated for most people
   b) They are always accurate because they show hard numbers
   c) People tend to trust them more than a plain claim, even though they can be manipulated
   d) They are only used in science, not in advertising
3. What is the speaker's overall advice at the end?
   a) Never trust any graph
   b) Only trust graphs from official organizations
   c) Look carefully at the labels, numbers, scale, and context, and ask what story the graph is telling
   d) Make your own graphs instead of using other people's
4. In your own words, name two ways a graph can be misleading. (Open answer)

Part 2: Cause-and-effect Language

Cause and effect is often described using conjunctions like *because*, prepositional expressions like *due to* and *because of* and adverbs like *therefore* and *consequently*. In this unit we focus on verbs and nouns relating to cause and effect. Pay particular attention to the prepositions they are used with.

A Verbs relating to cause and effect

You are probably already familiar with these verbs relating to cause and effect: *make, cause, create, do, produce, force*. Here are some other useful verbs.

Her grandmother **influenced/had a considerable influence on** Sarah's choice of career.

Parental attitude largely **determines** how well a child adapts to school. [is the main factor affecting it]

A good teacher **motivates** their students. [makes them want to do something positive]

The flow of traffic through the town is **facilitated** by the one-way system. [made easier]

His speech **provoked** an angry response. [caused, usually something negative]

The explosion was **triggered** by the heat. [started, usually something sudden and negative]

The tilting of the earth on its axis **accounts for** the change in the seasons. [explains]

Sid's determination **springs/stems from** his desire to improve the world. [is the result of]

The country's victory **gave rise to** a new mood in society.

Mobile technology **contributed to** the information revolution. [was one factor influencing]

Their child was given drugs to **stimulate** growth. [cause something to develop or function]

Her exhibition **generated** a lot of interest. [caused something to happen or exist]

The drugs may **induce** nausea. [cause, often used in a medical context]

A teacher's presence can often **inhibit** teenagers' discussions. [prevent them from being as free as they might otherwise have been]

A number of benefits can be **derived from** this situation. [gained as a result of something]

B Nouns relating to cause and effect

noun	example	meaning/comment
chain reaction	The incident set off a **chain reaction** which affected us all.	set of related events in which each one causes the next one
consequence	The war had major **consequences**.	results
effect, impact	The **effect/impact** of the film **on** the audience was very powerful	influence
end	Does the **end** justify the means?	note how end here means aim or goal
origin/source	The accident was the **origin/source of** her later problems.	beginning or cause
outcome	The **outcome of** the discussions was very positive.	result or effect of an action
precedent	There are several **precedents for** taking such a decision.	something that already happened and provides a reason for doing the same
reason	What was the **reason for** his success / why he succeeded?	note how reason is often followed by for or why

Practice Activities:

Activity 1

Choose the best verb to complete each sentence.

1 Researchers are investigating why chocolate *provokes* / *induces* headaches in certain people.
2 Wilson's most recent paper has *motivated* / *generated* a great deal of interest among sociologists.
3 The drug *triggered* / *facilitated* headaches and dizziness among a number of subjects in the test.
4 Having an end-of-term prize *contributes* / *motivates* students to do well in their class tests.
5 Intensive farming has *contributed to* / *provoked* the decline of bird populations.
6 The missile test *stimulated* / *provoked* an immediate and very strong response from neighbouring governments and from the United Nations.
7 Astrologists believe that people's lives are *motivated* / *determined* by the planets and stars.
8 The barrier *inhibits* / *induces* the flow of water into the area to prevent flooding in the rainy season.
9 A leak in the tank *stemmed from* / *accounted for* 40% of the fuel loss, and evaporation took away another 5%.
10 The renovated college buildings have wider doors to *influence* / *facilitate* the use of wheelchairs.
11 The Minister cut taxes in an attempt to *stimulate* / *trigger* the economy, which was performing poorly.
12 Britain has *facilitated* / *derived* many economic benefits from membership of the European Union.

Activity 2

Match the beginnings and endings of the sentences.

1	The article explores the origins	☐ a	major consequences for larger families.
2	One small explosion set off a chain	☐ b	precedents for his decision.
3	The confusion probably stemmed	☐ c	rise to a widespread sense of disillusionment.
4	The proposed new tax could have	☐ d	of the concept of democracy.
5	The disastrous events of 2003 gave	☐ e	reaction, causing massive damage.
6	The judge explained that there were	☐ f	from a lack of communication.

Activity 3

Complete the sentences with the missing words. In one sentence, it is not necessary to add a word.

1 De Routa's work in the 1970s influenced ..... the development of computer science.
2 The acid seemed to have no ..... whatsoever on the plastic, which remained unchanged.
3 Everyone hoped that the outcome ..... the meeting between the two governments would be a new and better trade agreement.
4 The reason ..... the failure of the project was a lack of funding.
5 The impact of global warming ..... the polar ice caps is now understood more clearly.
6 There is no precedent ..... this kind of decision.

Activity 4

Write the nouns related to these verbs. Use a dictionary if necessary.

motivate influence trigger contribute induce

Activity 5

Complete the sentences using the correct form of nouns from 37.4.

1 Byron's poetry had a significant ..... on the work of other Romantic poets.
2 Tree pollen can be a ..... for hay fever attacks in vulnerable people.
3 All the political parties illegally offered people ..... to vote for them, such as cash payments, free tickets for sports events, etc.
4 The ..... made by the burning of fossil fuels to global warming is now beyond dispute.
5 The ..... to get a good job and to travel and see the world is often more effective than the teacher or the teaching material in language learning.

Part 3: Listening to Short Talks – Structure and Signposting

When listening to a lecture, you need to be able to recognize the main points and supporting details. It is also important to understand how information and ideas are organized. Lecturers may use signposting expressions to show how their lecture will be organized - recognizing these expressions will help you understand what you are going to hear and the order in which you will hear it. Lectures also often include **definitions** and **explanations** of key terms and ideas. It is important to recognize these in order to take effective notes.



1 What motivates people?
2 How have theories developed?
3 How have they influenced each other?

Slide A

Theories of motivation

- Taylor
- Maslow
- McGregor
- McClelland & Burnham

Slide B

The history of motivational theories at work

Dr David Hughes
Institute of Psychology

Slide C

TASK 1 Using your knowledge to prepare for a lecture

1 Work in pairs. Discuss how the following factors are linked to motivation at work.

praise from your boss	earning a lot of money	your job title
being creative	having power	doing challenging work
respect from colleagues	a good pension	a sense of achievement

2 Which factors in 1 do you think are the most motivating?

TASK 2 Understanding the main points of a lecture

1 You are going to listen to extracts from a lecture about motivation at work. Look at slides A–C from the start of the lecture and decide which order they will appear in.

2 3.1 Watch Extract 1 and check your prediction in 1. Make brief notes on the main points the lecture will cover.

3 Work in pairs. Use your notes to tell each other what you understood from Extract 1.

TASK 3 Understanding the organization of a lecture

1 3.1 Watch Extract 1 again and complete the expressions in sentences 1–5 that show how the lecture will be organized.

1 _____ I'd like to look at a number of theories related to motivation ...

2 _____, we'll look _____ at Taylor's idea that money is the key motivator in the workplace.

3 Then, we'll _____ look at one of the most famous theories of motivation, which is Maslow's hierarchy of needs.

4 _____, we're _____ how Maslow's ideas influenced the work of Douglas McGregor ...

5 And _____, I want to _____ at a slightly different view of motivation developed by McClelland and Burnham ...

Language Reference

Signposting Language

Lecturers often use different expressions to show the organization of the lecture. For example:

Giving an overview

As we'll see from today's lecture ...
In today's lecture I'd like to look at ...
In this lecture, I'll be focusing on ...

Sequencing points

First, we'll look in more detail at ...
Then, we'll move on to look at ...
After that, we're going to see ...
And then finally, I want to take ...

TASK 4 Taking notes on key information

1 3.2 **Watch Extract 2 and label the timeline with the people behind the theories of motivation.**

1 McClelland and Burnham 2 Taylor 3 Maslow 4 McGregor



2 3.2 **Watch Extract 2 again and complete the notes.**

1 Taylor: _____ motivates everyone

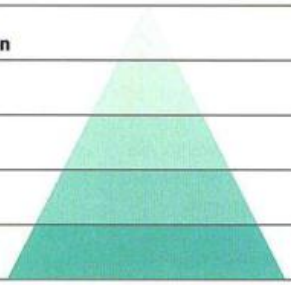
2 Maslow: motivation = _____ main areas

3 McGregor: management styles = _____ groups: _____ and _____

4 McClelland and Burnham: studied _____ main motivator = _____

TASK 5 Taking detailed notes on explanations and examples

1 3.3 **Watch Extract 3 of the lecture. Complete the notes on each level of Maslow's hierarchy of needs.**



	Explanation	Example related to work
Self-actualization	7 _____	8 _____
Esteem	*Status – being accepted and respected*	6 _____
Belonging	4 _____	5 _____
Security	2 _____	*Job security and* 3 _____
Physiological	*Basic survival needs – food, water, etc.*	1 _____

2 3.3 **Watch the last part of Extract 3 again and answer questions 1–4.**

1 Which need is fulfilled first?

2 Can you move from security needs straight to esteem needs?

3 What happens when one need is met?

4 Does everyone have to reach the top of the pyramid to be satisfied?

3 Compare your answers in 1 and 2 with another student. Help each other to complete any missing information.

TASK 6 Recognizing phrases for signposting a lecture

1 Match the following phrases the lecturer uses with functions a–c.

As I said So for example this can be understood as In other words
An obvious example of this is How that works is As we've already seen

a giving an example b giving an explanation c referring to something said earlier

2 3.3 **Watch Extract 3 again and check your answers.**